A Multitask Network and Two Large-Scale Datasets for Change Detection and Captioning in Remote Sensing Images

Jingye Shi, Mengge Zhang, Yuewu Hou, Ruicong Zhi, and Jiqiang Liu

***Abstract*— Remote sensing change detection (RSCD) recognizes pixel-level change regions between images, while remote sensing change captioning (RSCC) describes the nature and properties of these changes in natural language. The past have studied two tasks individually. Resulting in a single interpretation content produced and limited application scenarios. And the complementarity of change regions and deep semantic information can further improve the performance and robustness of the model. Therefore, we try simultaneously to solve both RSCD and RSCC tasks (i.e., RS-CDC) under a multitask framework, namely a CNN-Transformer-based multitask network (CTMTNet). Specifically, we design multiattention feature enhancement module (MAFEM) and feature fusion block (FFB) to enhance local information and location perception of features from bitemporal images. The MAFEM weights the channel and space separately to capture local information more accurately and enhance location perception. The FFB fuses bitemporal features and uses multilevel residual connections to ensure that change information is not lost during transfer. Finally, we use two decoders to output the change maps (CMs) and change captioning, respectively. During training, we use an improved multitask loss function for CTMTNet to balance the two tasks. For exploring the RS-CDC task, we construct two large-scale datasets named LEVIR-CDC and WHU-CDC dataset. We benchmark the existing state-of-the-art (SOTA) change detection (CD) and change captioning methods on these two datasets and a newly publicized LEVIR-MCI dataset, and the results show that the proposed CTMTNet significantly outperforms comparative methods.**

***Index Terms*— Attention mechanism, change captioning, change detection (CD), multitask learning, remote sensing (RS) images.**

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Jingye Shi and Jiqiang Liu are with Beijing Key Laboratory of Security and Privacy in Intelligent Transportation, Beijing Jiaotong University, Beijing 100044, China (e-mail: shijy@geovis.com.cn; jqliu2015@126.com).

Mengge Zhang is with the National Supercomputing Center and the School of Computer and Artificial Intelligence, University of Zhengzhou, Zhengzhou 450001, China (e-mail: zmg080605@163.com).

Yuewu Hou and Ruicong Zhi are with the School of Computer and Communication Engineering and Beijing Key Laboratory of Knowledge Engineering for Material Science, University of Science and Technology Beijing, Beijing 100083, China (e-mail: 1078332181@qq.com; zhirc@ustb.edu.cn).

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I. Introduction

In RECENT years, owing to the ever-expanding range of human activities and the evolution of the natural environment, the surface of the Earth has changed dramatically. The rapid development of remote sensing (RS) sensor technology has promoted the wide application of RS technology in the field of Earth observation, including global resource monitoring [1], land use change detecting [2], [3], urban management [4], [5], and environmental protection [6], [7]. At the same time, many intelligent interpretation tasks have emerged, such as object detection [8], [9], image classification [10], RS change detection (RSCD), and RS change captioning (RSCC). Among these, RSCD can be further divided into RS binary change detection (RSBCD) and RS semantic change detection (RSSCD), which, along with RSCC, serve to interpret changes on the Earth's surface.

RSBCD is the most basic type of change detection (CD), aiming to determine whether changes exist between two temporal images. It typically generates a binary change mask that clearly identifies where changes have occurred. In contrast, RSSCD adds information about land cover categories to this. It usually combines semantic segmentation tasks with BCD tasks, as in [11] and [12], using a multitask network (containing two semantic segmentation decoders and one CD decoder) to simultaneously perform category segmentation of bitemporal images and generate a change mask. By combining the CD map with the two category segmentation maps, two change segmentation maps (corresponding to the images before and after the change) can be obtained. RSCC integrates techniques from computer vision (CV) and natural language processing (NLP) to describe changes in bitemporal images using natural language. Compared to RSSCD, RSCC contains richer and more diverse semantic information, such as the nature, properties, location, and relative position of changes, thus providing more varied and detailed interpretative results.

The RSCD and RSCC tasks are both important tools for analyzing change information in RS images, and significant advances have been made in these areas in recent years. However, previous research has largely treated these two tasks as independent and trained them separately. In practical applications, the analysis results from a single task may not comprehensively capture all aspects of changes, thereby limiting their scope of application. Moreover, training a robust model individually is costly, as it requires substantial time and

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effort for model optimization and hyperparameter tuning, and handling the two tasks separately is even more time-consuming and labor-intensive. More importantly, a single model overlooks the potential connections and complementarity between RSCD and RSCC. For example, the positional information provided by RSCD can improve the accuracy of location attributes in the captions generated by RSCC; conversely, the additional semantic supervision signals provided by RSCC can enhance RSCD's understanding of key features of changed objects. Therefore, we consider adopting a multitask learning approach that combines RSBCD (hereafter referred to as RSCD) with RSCC, forming the RS-CDC task. Within a unified framework, this approach achieves both tasks in a more concise and efficient manner, obtaining richer interpretive results. During training, utilizing more supervisory signals from both tasks enables the model to learn more general and robust feature representations, thereby enhancing its performance and robustness.

To explore the RS-CDC task and verify our approach, we build two large-scale RS-CDC datasets named LEVIR-CDC and WHU-CDC dataset, respectively. We implement the RS-CDC task in a multitask framework and propose a CNN-Transformer-based multitask network (CTMTNet). To reduce the number of total parameters of the model, and one task can utilize the features of the other, we let the two tasks share the feature extraction backbone (CNN-based weight-sharing Siamese network) and a new feature enhancement and fusion module (FEFM). The FEFM consists of a multiattention feature enhancement module (MAFEM) and a feature fusion block (FFB). The MAFEM uses channel and spatial attention (SA) mechanism [13] to reweight channels and space to capture local information more accurately and enhance location perception, while the FFB uses convolution operations to fuse bitemporal features. The FFB also uses a multilevel residual connection to ensure that change information is not lost during the transfer process. Finally, we designed independent decoders for the two tasks to form a multitask decoder that acquires the change maps (CMs) and the sentences describing the changes, respectively. In the training phase, we perform supervised training using an improved multitask loss function to balance the two tasks. We have conducted extensive experiments on the two proposed datasets and a newly publicized LEVIR-MCI dataset, and the results show that our proposed method has higher performance.

The main contributions of this article are as follows.

1) We employed a unified framework that efficiently integrates the RSCD and RSCC tasks (RS-CDC), achieving complementary utilization of positional and semantic information, which in turn enhances the model's performance and robustness. This significantly saves time and reduces the workload compared to handling the two tasks independently.
2) For the RS-CDC task, we propose a novel multitask model, CTMTNet, which can provide richer interpretation results. The two tasks learn shared representations at the lower levels of the model and generate corresponding outputs through independent CD and CC decoders. In the training phase, we employ an improved multitask loss-weighting method to balance the learning speeds of the two tasks, which further improves the performance of the overall model. Extensive experiments demonstrate the effectiveness of the method.
3) We have proposed two large-scale datasets named LEVIR-CDC and WHU-CDC dataset to further the development of the RS-CDC task. We benchmarked existing state-of-the-art (SOTA) CD and CC task approaches on these two datasets and a newly publicized LEVIR-MCI dataset, respectively. Our CTMTNet outperforms these approaches.

The other sections of this article are organization as follows. Section II reviews related work. Section III details the two proposed datasets. Section IV details the proposed method. Experimental data, evaluation metrics, and training details are discussed in Section V. Section VI provides discussion and conclusions.

II. Related Works

RSCD, RSCC, and multitask learning are the foundational research areas of our work. Next, we will provide a brief overview of these three research topics.

A. RS Change Detection

RSCD involves identifying significant changes between RS images. Compared with natural images, RS images have lighting variations and registration errors, and the scenes become more complex.

For CD methods, the earlier ones mainly extracted change regions by threshold segmentation of pixel differences to generate CMs, including algebraic operations [14], [15], [16], [17] and image transformation [18], [19]. However, these methods lacked neighboring pixel and spectral information and were susceptible to noise interference. As a result, some salt-and-pepper noise was easily mixed in the final obtained CMs. Subsequently, machine learning (ML) methods emerged, including support vector machines (SVMs) [20] and artificial neural networks (ANNs) [21]. ML approaches utilized existing datasets to train classification models using algorithms (e.g., decision trees and SVM) to distinguish between changed and unchanged regions. It has improved performance compared with earlier methods, but often lacks enough semantic information to perform well in semantic-rich scenes. Today, with the development of RS technology, the acquired images contain increasingly rich information, and scenes are becoming more complex. The aforementioned methods often struggle to deliver optimal performance under these conditions. The emergence of DL-based methods has brought a breakthrough in the RSCD task.

In contemporary research, most of the DL-based approaches are designing Siamese networks and using pre-trained backbone networks (e.g., ResNet, VGG, etc.) for extracting high-level semantic features from bitemporal RS images. Subsequently, a decoder is utilized to fuse bitemporal features for identifying change regions in complex scenes. For example, to capture multilevel features from bitemporal images and enhance CD performance, Chen and Shi [4] designed a

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Siamese attention network. RS images also have the problems of change objects with different sizes. Consequently, researchers have proposed many multiscale feature aggregation strategies. For example, Wang et al. [22] used a Siamese full convolution as an encoder and reconstructed CMs for different levels of features during the decoding stage. Fang et al. [23] have used compact and dense connections between encoders and decoders, as well as between decoders, to improve the problems of small target missing and edge details of changed objects. The spatio-temporal modeling approach enhanced the precise identification of change regions while solving traditional convolutional neural networks (CNNs)' limitations in capturing global contextual information and long-term dependencies. For example, Yan and Jiang [24] incorporated spatio-temporal domain modeling within the context using a transformer encoder and utilized the transformer as a decoder to refine the original features. Additionally, Bandara and Patel [25] effectively represented multiscale remote details by combining transformer and multilayer perceptron (MLP). To enhance the applicability of algorithms to real-world scenarios, researchers have focused on developing lightweight and efficient CD algorithms. For example, Li et al. [26] designed a lightweight network based on mobile networks and a neighbor aggregation module to compensate for weak representations. Zhang et al. [27] introduced an asymmetric multihead cross-attention module to boost interaction between CNNs and features extracted by transformers, simultaneously reducing the transformer's quadratic computational complexity to linear.

B. RS Change Captioning

The RSCC task is a new topic in the interdisciplinary field of CV and NLP. The RSCC task originates from the image captioning (IC) task. Currently, most of IC methods are based on encoder–decoder approaches [28], [29], [30]. Typically, visual encoders or CNNs are used to extract image features, while language models serve as decoders (such as SVMs, recurrent neural networks (RNNs), and Transformers) to generate captions, which provides inspiration for the RSCC task. However, unlike the IC task, which only deals an image, the RSCC task needs to handle bitemporal or multitemporal images, analyze changes between the images, and describe them using natural language.

Currently, RSCC primarily employs deep learning methods. Initially, Chouaf et al. [31] used a pre-trained CNN as an encoder and simply concatenated the features of bitemporal images to obtain a fused feature, finally used an RNN as a decoder to obtain change captions. Later, Hoxha et al. [32] utilized two fusion strategies (early fusion and late fusion) to integrate bitemporal features, along with two decoders (RNN and k-class SVM) to obtain change captions. In the same year, Liu et al. [33] proposed a large-scale dataset called LEVIR-CC and the RSICCformer model. The RSICCformer used a CNN to extract visual features and designed a differential module and a fusion module, which were used to extract and integrate visual change information, respectively. A Transformer decoder then integrated multimodal information to generate corresponding change descriptions.

However, RS images contain a large number of objects with different scales, and current methods cannot adequately extract and utilize multiscale information. PSNet [34] used a transformer-based progressive scale-aware network to fully exploit multiscale information for generating change descriptions. ICT-Net [35] used a CNN to extract multiscale features and designed attention encoders that focus on differences and content separately, obtaining multilevel change-aware representations. By connecting the bitemporal change-aware representations, it used a gated attention mechanism to obtain change representations, and finally, it used a decoder based on the transformer to generate change captions. The above approaches ignore the fact that the difficulty of RSCC differs for changing and non-changing image pairs. To this end, PromptCC [36] decoupled the RSCC task into two questions: whether there was a change and what kind of change has occurred. They introduced a pre-trained large language model into the RSCC task and utilized its powerful ability to generate accurate descriptions.

C. Multitask Learning

Multitask learning is usually a primary task combined with one or more subsidiary tasks. The results of many tasks influence each other and require methods to balance them, to avoid being dominated by one task. There are three commonly used approaches: first, designing network architectures that utilize task hierarchies, feature fusion, and attention mechanisms to design multitask network architectures, such as [37] and [38]. Second, the loss functions of multitasks use different weighting methods, so that the optimum loss weights are used for each epoch during the training process, as shown in [37], [39], [40], and [41]. Third, the effect of multitask learning is enhanced by resolving the gradient conflicts that arise during the training process and reducing the conflicts between different tasks, as illustrated in [42], [43], [44], [45], and [46].

There were two main approaches to network architecture design in deep learning, hard parameter sharing (HPS), and soft parameter sharing (SPS). HPS has emphasized that all tasks share the hidden layer, and each task corresponds to a sub-network, such as [37]. It reduced the number of total model parameters and lowered the risk of overfitting. In the case of limited data, the knowledge learned by one task could be passed on to other tasks, improving the overall model performance. Comparatively, when the complexity and size between tasks were inconsistent, the shared network might be biased toward learning task features that were easy to converge or data-rich, failing to capture task-unique features and leading to an imbalance. Each task in SPS has a specialized network layer that uses regularization and other methods to keep the parameters between tasks similar, such as [38]. It enabled the model to capture both the shared features between tasks and the unique features of the tasks, and could handle the condition of large differences between tasks more effectively. However, the parameter number of the model would increase, requiring more computational resources and training time, and it was not easy to adjust the methods such as regularization, which required the choice of suitable hyperparameters.

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When specific weights were needed to combine the loss functions of multitasks, a manual weighting method that the weights of the loss functions of each task were set manually was initially used. But the method was task-intensive. Therefore, researchers have proposed a number of dynamic weighting methods that adaptively adjust the weights of the loss functions during the training phase. For example, Liu et al. [37] considered both labeling loss and gradient loss in order to mitigate the effect of differences in the magnitude of different losses, while expressing the gradient as a function of the weights to control the training of the network. Based on [37], Chen et al. [39] utilized the change in loss to measure the learning progress of a task and used the rate of change of loss to dynamically adjust the weights of the multitasks. Kendall et al. [41] proposed using the same variance uncertainty to dynamically balance feature task loss functions weights in multitask loss, giving higher weights to "simple" tasks. These methods do not consider the correlation between tasks, and when there is little correlation between tasks, gradient conflicts may occur. Therefore, Liu et al. [40] proposed Auto-λ, which explicitly encodes the relationship between tasks using a set of task weights related to training loss and updates these task weights dynamically as the multitask network parameters are trained.

When tasks compete with each other, it was not enough to use only the first two methods. Therefore, researchers have proposed a number of methods for resolving gradient conflicts that arise during training. For example, Sener and Koltun [42] equated multitask learning to a multiobjective optimization problem for finding the optimal solution by obtaining the weights corresponding to each task in a common gradient descent direction to find the Pareto optimal solution. Based on [42], Lin et al. [43] proposed a new Pareto multitask learning method to find a set of well-distributed Pareto optimal solutions. These methods solved the gradient conflict by Pareto optimal solution. In addition to this, some researchers have proposed to mitigate the gradient conflict problem through gradient update directions. Yu et al. [44] eliminated the gradient update conflict between two tasks by projecting the gradient of the other task on the normal vector by one task. Based on [44], Wang et al. [45] set the minimum acceptable value of the cosine of the pinch angle based on the relationship between the different tasks to identify and eliminate conflicts. Liu et al. [46] were avoiding gradient conflicts by solving the gradient updater in a neighborhood of the average gradient direction.

III. Proposed Datasets

In recent years, the RS-CDC task has just started, with fewer related studies and a lack of unified evaluation datasets and metrics. Therefore, this article proposes two new datasets that are suitable for RS-CDC, namely LEVIR-CDC dataset and WHU-CDC dataset. Datasets available at https://huggingface.co/datasets/hygge10111/RS-CDC.

As far as we know, these two datasets address both RSCD and RSCC for the first time. They are built based on existing datasets that reference the datasets LEVIR-CC [33] and WHU-CD [47]. The LEVIR-CC dataset is already labeled with the change captions in the bitemporal RS images, and we supplement the binary change labels based on them. The WHU-CD dataset already contains binary change labels, and we supplement the change captions of the bitemporal RS images. In the following, we describe each dataset and the production process.

A. LEVIR-CDC Dataset

The LEVIR-CC dataset was based on the LEVIR-CD dataset [4], which contains 637 high-resolution (0.50 m/pixel) bitemporal RS images and each image has a spatial size of 1024 $\times$ 1024 pixels. These bitemporal images originate from 20 regions in Texas, USA, spanning a time period of 5–15 years. To be suitable for the task of RSCC, the images of the LEVIR-CD dataset have been cropped to a size of 256 $\times$ 256 pixels, and supplement some images to maintain a balanced number of change and unchanged image pairs. Therefore, the LEVIR-CC dataset contains 10 077 pairs of images consisting of 5038 changed image pairs and 5039 unchanged image pairs. Each pair of images has five annotated sentences to describe the changes between bitemporal images. The maximum length of the sentence was 39 words, while the average is 7.99 words. Liu et al. [33] provided a detailed description of the LEVIR-CC dataset.

In order to make the dataset suitable for the RS-CDC task, we supplement binary change labels for the bitemporal images based on the LEVIR-CC dataset. First, we crop the bitemporal images in LEVIR-CD to a size of 256 $\times$ 256 pixels based on the cropping method of the LEVIR-CC dataset. Then, we obtain the corresponding binary change labels. Finally, for supplementary those image pairs, we use labelme (https://github.com/wkentaro/labelme), an RS image labeling tool, to mark the change regions. Ultimately, this dataset contains 10 077 image pairs, 10 077 labels, and 50 385 annotated sentences describing the changes. Fig. 1(a) shows an example from the dataset.

Recently, Change-Agent [48] proposed a new dataset called LEVIR-MCI that contains both CM and caption labels. The LEVIR-CDC and LEVIR-MCI are both extensions of the LEVIR-CC dataset that supplement the labels of the CD task. We compared the two datasets at the directory level, as shown in Table I. A major difference between the two datasets is the change categories contained in the CM labels of the image pairs. The former uses a binary change label that contains one change category (i.e., buildings), while the latter uses a multicategory change label that contains two change categories (roads and buildings), and each is represented by a different color.

B. WHU-CDC Dataset

The images in the WHU-CDC dataset are primarily from the CD dataset WHU-CD [47] and consist of high-resolution (0.075 m) image pairs of 32 507 $\times$ 15 354 pixels. It recorded the construction of buildings in the Christchurch region following the 2011 February 6.3 magnitude earthquake in New Zealand.

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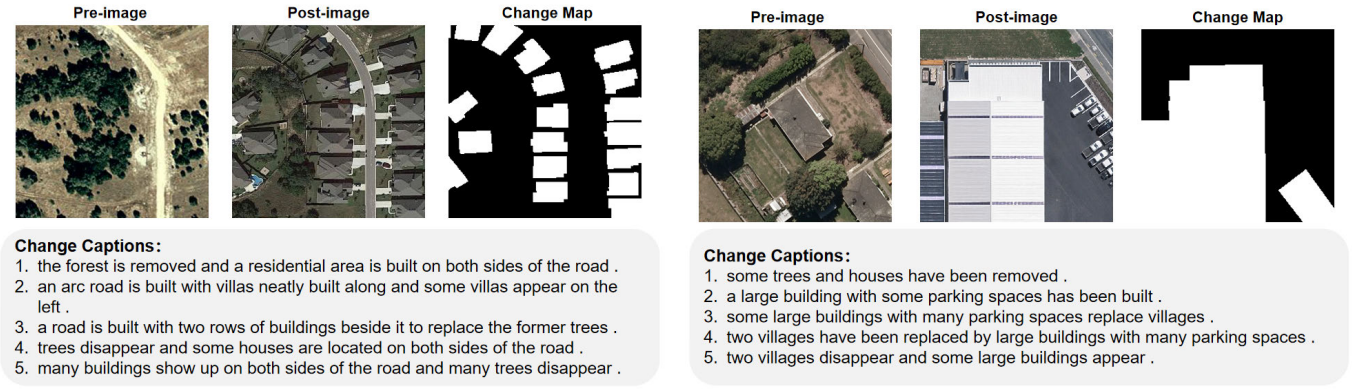


Fig. 1. CM examples and reference captioning from (a) LEVIR-CDC and (b) WHU-CDC datasets, respectively.

TABLE I

COMPARISON OF THE LEVIR-CDC AND LEVIR-MCI DATASETS. SIZE MEANS THE IMAGE SIZE. A, B, AND CMS DENOTE THE PRE-IMAGE, POST-IMAGE, AND CORRESPONDING CHANGE LABELS IN THE THREE SETS (TRAIN, VAL, AND TEST), RESPECTIVELY. THE CAPTIONS.JSON FILE HOLDS THE CC LABELS OF ALL IMAGE PAIRS

		Size	Training/Validation/Test			captions.json
			A	B	CMs	
Identical		✔	✔	✔	✘	✔
Differences	CDC	-	-	-	building	-
	MCI	-	-	-	building,road	-

TABLE II

PERCENTAGE OF CHANGED OR UNCHANGED IMAGE PAIRS IN TRAINING, VALIDATION, AND TEST IN THE WHU-CDC DATASET

Change or not	Training(%)	Validation(%)	Test(%)
Change	79.89	10.50	9.61
No Change	80.05	9.74	10.20
Total	80.00	9.99	10.01

TABLE III

NUMBER OF IMAGE PAIRS PER CHANGE CATEGORY IN TRAINING, VALIDATION, AND TEST IN THE WHU-CDC DATASET. WHERE M.C. MEANS THE AVERAGE NUMBER OF CHANGES IN EACH IMAGE PAIR

Cetegory	Training	Validation	Test	Total	Proportion(%)
Building	1447	193	168	1808	73.62
Road	1183	162	146	1491	60.71
Parking lot	236	25	32	293	11.93
Vegetation	499	63	59	621	25.29
Water	27	8	3	38	1.55
M.C.	1.73	1.75	1.73	1.73	-

The WHU-CDC dataset contains 7434 image pairs of size 256 × 256 pixels. We collected five annotated sentences for each image pair describing the changes between RS images, which were provided by five different annotators with a background in RS. During the annotation process, first the scenes in the bitemporal image were visually inspected to determine the nature of the changes, considering attributes (i.e., color and size) as well as the relationship, location, and relative position to changes in other objects. Then, annotators need to annotate according to annotation rules [33]. We collected all the annotated sentences, checked and modified the sentences for spelling and grammatical errors. Finally, our dataset contains 37 170 sentences, and Fig. 1(b) shows an example from the dataset.

In Table II, we report the percentage of image pairs that have changes and have no changes in training, validation, and test. We focused on five object categories of changes: 1) building-related changes; 2) parking lot-related changes; 3) road-related changes; 4) vegetation-related changes; and 5) water-related changes. As shown in Table III, there are statistics on the number of each category in training, validation, and test for each image pair. It can be noticed that the proportions are almost the same in all three sets, both in the percentage of changes and no changes and the number of change categories, indicating that the distribution of image pairs is similar in training, validation, and test. The average number of changes for each image pair in the three sets is similar or consistent with that of the total dataset. We also calculated the percentage of sentences of different lengths, as shown in Fig. 2. We can see that the distribution of sentence lengths is similar in training, validation, and test. In the WHU-CDC dataset, the minimum sentence length is three words, and the maximum length is 24 words. The total vocabulary size of the dataset is 327 unique words.

IV. METHODOLOGY

In this section, we first give an overview of the proposed method, then the FEFM and the multitasking decoder, respectively, and finally, we describe the multitasking loss used by the proposed method.

A. Overview

In this work, we design a new multitask network (CTMTNet) for the RS-CDC task. The two tasks share the hidden layer and only separately handle the shared features in the final decoding stage for obtaining the corresponding results. Compared to handling the two tasks independently, we concisely and efficiently implement both RSCD and RSCC tasks simultaneously. This significantly saves time and reduces the workload. The overall architecture is shown in Fig. 3. The

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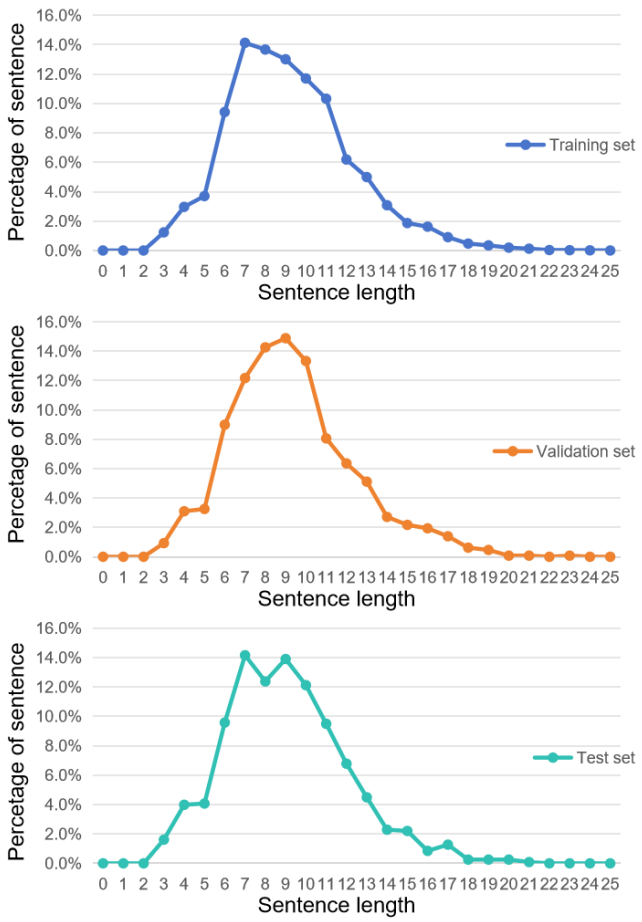


Fig. 2. Percentage of sentences of different lengths in training, validation, and test.

CTMTNet contains a CNN-based feature extractor, an FEFM module for feature enhancement and fusion, and a multitask decoder.

The CTMTNet takes aligned bitemporal RS images (T_1 and T_2) as input. The feature extractor uses ResNet101 [49] as the backbone to extract high-level features (X_{T_1} and X_{T_2}) from bitemporal RS images. And positional embeddings are added to this features to assign a unique representation to each position. X_{T_1} and X_{T_2} contain rich semantic information and more abstract representations, but lacks local information and weak location perception, which is more favorable for the RSCC task. However, the RSCD task requires sufficient local information and strong location perception to acquire small changes in the image and accurately localize the boundaries and outlines of the changed objects. To compensate this shortcoming, we introduce the FEFM module, which contains MAFEM and FFB sub-modules. The MAFEM is used to enhance the focus on local information and improve the perception of the object. The FFB module fuses the enhanced bitemporal features. It uses a residual block to reduce computational complexity and improve the ability to capture local and location information, resulting in improved perception of changing object locations. Finally, the fused feature F'_{fus} is input to the multitask decoder. It contains a CD decoder and a CC decoder. The former outputs a binary CMs by using F'_{fus}. The latter uses F'_{fus} and word embeddings to output a sentence that describes the difference between the bitemporal RS images. The CTMTNet uses the same input (i.e., bitemporal RS images) to accomplish the RSCD and RSCC tasks concisely and effectively at the same time, and the results of the experimental comparisons with the SOTA methods also verify the advancement and effectiveness of the CTMTNet.

B. Feature Extractor

Currently, the extraction of high-level features from bitemporal RS images $T_1 \in \mathbb{R}^{H\times W\times 3}$ and $T_2 \in \mathbb{R}^{H\times W\times 3}$ is generally done using CNN-based Siamese networks, such as ResNet18 [49], ResNet101 [49], AlexNet [50], GoogLeNet [51], and VGG [52]. In our work, the ResNet101 network pre-trained on the ImageNet dataset [53] is used as the backbone. Specifically, we feed the image pair T_1 and T_2 into the ResNet101 network to obtain high-level features, which is represented as follows:

$$X_{T_i} = f_{\text{CNN}}(T_i), \quad (i = 1, 2) \tag{1}$$

where i represents the temporal dimension of image pairs, f_{CNN} denotes the function of the feature extractor, $X_{T_i} \in \mathbb{R}^{H/16\times W/16\times C}$ denotes the high-level features, where H, W, and C denote the height, width, and channel of images. We can effectively use a Siamese backbone with weight sharing to extract high-level features from the bitemporal RS images as the foundation for our subsequent stages.

C. Feature Enhancement and Fusion Module

The FEFM contains two sub-modules, MAFEM and FFB. We will describe these two modules in detail below.

1) Multiattention Feature Enhancement Module: In order to enhance the ability of the model to extract local information and location perception, we design the MAFEM module, which is shown in Fig. 4. In MAFEM, we introduce the channel attention (CA) mechanism and the SA mechanism. The image features are compressed in spatial dimension using the CA mechanism to get a 1-D vector for channel weighting. It enables the model to focus on local features that are highly correlated with changes in the channel dimension. The SA mechanism is used to compute weights for each location of the image features to determine which location should be highlighted and which ones should be ignored. It enables the model to capture the local information of changing objects more accurately and improves the location perception. The reweighted features are passed to a fully connected feed forward (FF) network, and residual connection and normalization operations to make the model more stable in learning a valid feature representation.

The MAFEM module accepts the feature with added position embedding (PE) as the initial input

$$X'_{T_i} = X_{T_i} + X_{\text{pos}}, \quad (i = 1, 2) \tag{2}$$

where $X_{\text{pos}} \in \mathbb{R}^{H/16\times W/16\times C}$ is a learnable 2-D PE that incorporates spatial location information into a bitemporal

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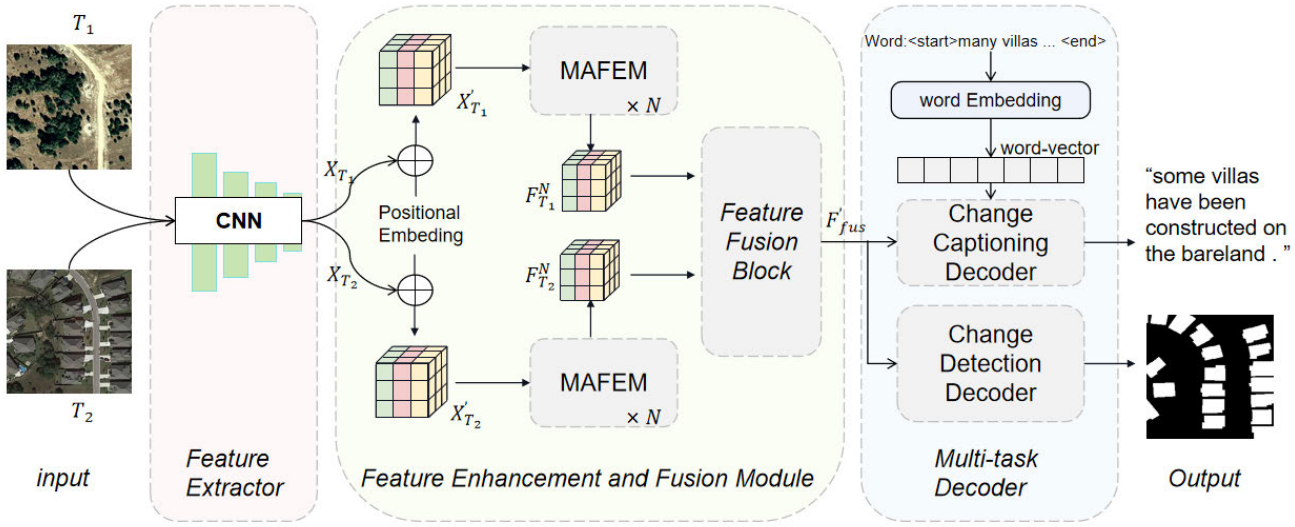


Fig. 3. Overview of the proposed multitask network (CTMTNet) with a shared backbone for feature extraction, a middle module for feature enhancement and fusion, a multitask decoder for generating CD, and change captioning.

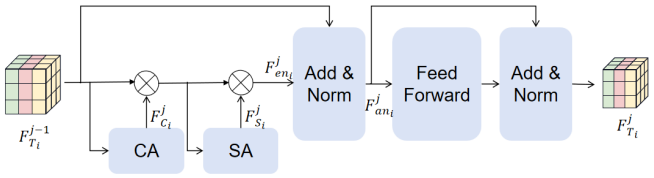


Fig. 4. Multiattention feature enhancement module.

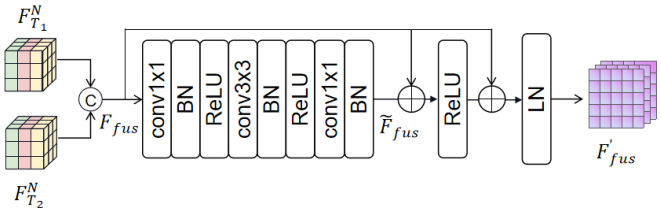


Fig. 5. Feature fusion block.

feature sequence [54], and $X'_{T_i} \in \mathbb{R}^{H/16\times W/16\times C}$ represents visual features with added PE.

Then, the dimension of feature X'_{T_i} is converted to $hw \times C$ ($h = H/16, w = W/16$) as the input $F^0_{T_i}$ for the MAFEM. The MAFEM contains N blocks, and the jth block can be represented as

$$F^j_{C_i} = \mathrm{CA}^j\left(F^{j-1}_{T_i}\right), \quad (i = 1, 2) \tag{3}$$

$$F^j_{S_i} = \mathrm{SA}^j\left(F^{j-1}_{T_i} \otimes F^j_{C_i}\right), \quad (i = 1, 2) \tag{4}$$

$$F^j_{\mathrm{en}_i} = F^{j-1}_{T_i} \otimes F^j_{C_i} \otimes F^j_{S_i}, \quad (i = 1, 2) \tag{5}$$

$$F^j_{\mathrm{an}_i} = \mathrm{LN}\left(F^j_{\mathrm{en}_i} + F^{j-1}_{T_i}\right), \quad (i = 1, 2) \tag{6}$$

$$F^j_{T_i} = \mathrm{LN}\left(\mathrm{FF}\left(F^j_{\mathrm{an}_i}\right) + F^j_{\mathrm{an}_i}\right) \tag{7}$$

where $j = 1, \ldots, N$ represents the depth of the MAFEM module, and LN represents layer normalization. The fully connected FF layer is composed of two linear layers and a rectified linear unit (ReLU) activation function. CA^j and SA^j represent CA and SA, respectively.

2) Feature Fusion Block: In order to fuse the bitemporal image features and extract the change information, we designed an FFB module (as shown in Fig. 5). In the FFB module, we first concatenate the enhanced bitemporal features and then input them into a residual block. The residual block contains three 2-D convolutional layers following a ReLU activation function. To ensure that change information is not lost during the transfer process, we use a multilevel residual connection. Finally, to make the obtained features better adapted to the decoders of both tasks, a layer normalization (LN) is used in the FFB module to normalize the distribution of the features. The computation of FFB can be expressed as follows:

$$F_{\mathrm{fus}} = \left[F^N_{T_1}; F^N_{T_2}\right] \tag{8}$$

$$\tilde{F}_{\mathrm{fus}} = \mathrm{Conv}_3(\mathrm{ReLU}(\mathrm{Conv}_2(\mathrm{ReLU}(\mathrm{Conv}_1(F_{\mathrm{fus}}))))) \tag{9}$$

$$F'_{\mathrm{fus}} = \mathrm{LN}(\mathrm{ReLU}(\tilde{F}_{\mathrm{fus}} + F_{\mathrm{fus}}) + F_{\mathrm{fus}}) \tag{10}$$

where $F^N_{T_i}(i = 1, 2)$ is the enhanced feature obtained from the MAFEM. [;] denotes a connection operation. Conv1, Conv2, and Conv3 are three 2-D convolutional layers with kernel sizes of 1×1, 3×3, and 1×1, respectively. LN represents layer normalization. The final output is the fused features F'_{fus}.

D. Multitask Decoders

In order to realize the output of CMs and change caption at the same time, we design two prediction headers in the multitasking decoder, namely CC decoder and CD decoder.

1) Change Captioning Decoder: In recent years, transformer has been widely used in the fields of CV and NLP, and both have achieved good results. Therefore, we use transformer as a CD decoder to obtain change descriptions. As shown in Fig. 6, the decoder consists of multiple transformer layers, each containing a masked multihead attention

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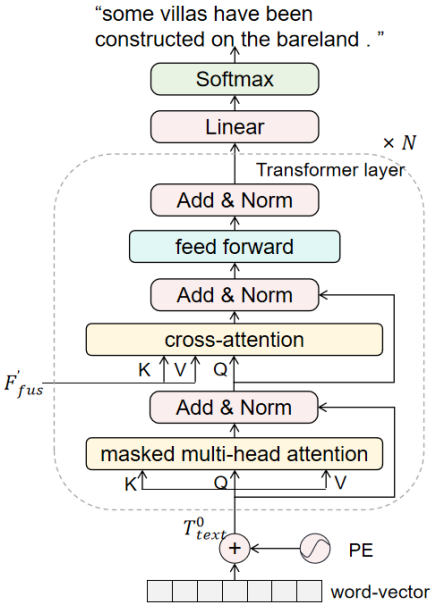


Fig. 6. Change captioning decoder.

sublayers (MHASs), a cross-attention sublayers (CASs), and a feed-forward sublayers (FFSs). Residual connections and LN operations are used between sublayers. Finally, a linear layer and a softmax activation function generate outputs.

First, each token (t) of the input sequence is mapped into a vector representation and added with PE as the initial input T^0_{text} of multiple transformer layers, denoted as follows:

$$T^0_{\text{text}} = f_{\text{vector}}(t) + T_{\text{pos}} \tag{11}$$

where T_{pos} is the PE calculated based on a sine function of the token's position, as described in [55] and [56].

In the transformer layer, MHAS uses a low triangle mask with the same shape as the attention weight matrix so that the model does not focus on the ungenerated parts of the output sequence to avoid information leakage. The CAS accepts the MHAS output as query and the visual features from the FEFM module as key and value to process the multimodal features. The results of the transformer layer are then output by FFS. The jth transformer layer can mathematically be expressed as follows:

$$W^j_{\text{text}} = \text{MHAS}^j\left(T^{j-1}_{\text{text}}, T^{j-1}_{\text{text}}, T^{j-1}_{\text{text}}\right) \tag{12}$$

$$W^j_{\text{text-img}} = \text{CAS}^j\left(W^j_{\text{text}}, F'_{\text{fus}}, F'_{\text{fus}}\right) \tag{13}$$

$$T^j_{\text{text}} = \text{FFS}^j\left(W^j_{\text{text-img}}\right) \tag{14}$$

where MHAS^j, CAS^j, and $\text{FFS}^j (j = 1, \ldots, N)$ represent the function of the jth MHAS, CAS, and FFS, respectively.

After the N-layer transformer, the word vectors retrieved are fed into the linear layer and softmax activation function to get the results.

2) CD Decoder: We design a lightweight decoder to predict the CMs. As shown in Fig. 7, we first perform chunking on the fused features $F'_{\text{fus}} \in \mathbb{R}^{H/16\times W/16\times 2C}$ obtained from the FEFM module to obtain bitemporal visual features $F'_{T_1} \in \mathbb{R}^{H/16\times W/16\times C}$ and $F'_{T_2} \in \mathbb{R}^{H/16\times W/16\times C}$. Then, the difference

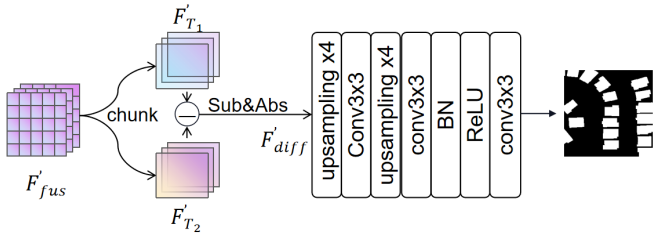


Fig. 7. CD decoder.

features $F'_{\text{diff}} \in \mathbb{R}^{H/16\times W/16\times C}$ are obtained by subtraction and absolute value operations. We upsample F'_{diff} to a size of $H/4 \times W/4$ and reduce the channel size through a 3×3 convolution, obtaining $\hat{F}'_{\text{diff}} \in \mathbb{R}^{H/4\times W/4\times C/16}$. Then, we upsample $\hat{F}'_{\text{diff}}$ to a size of $H \times W$ and obtain $\tilde{F}'_{\text{diff}} \in \mathbb{R}^{H\times W\times C/16}$. Finally, we use two 3×3 convolutions and a ReLU activation function to predict the CMs ($\text{CM} \in \mathbb{R}^{H\times W\times N}$), where $N(=2)$ represents both changed and unchanged classes. This process can be expressed as follows:

$$F'_{T_1}, F'_{T_2} = \text{Chunk}\left(F'_{\text{fus}}\right) \tag{15}$$

$$F'_{\text{diff}} = \text{Abs}\left(F'_{T_2} - F'_{T_1}\right) \tag{16}$$

$$\hat{F}'_{\text{diff}} = \text{Conv}_{3\times 3}\left(\text{Upsampling}\left(F'_{\text{diff}}\right)\right) \tag{17}$$

$$\tilde{F}'_{\text{diff}} = \text{Upsampling}\left(\hat{F}'_{\text{diff}}\right) \tag{18}$$

$$\text{CM} = \text{Conv}_{3\times 3}\left(\text{ReLU}\left(\text{Conv}_{3\times 3}\left(\tilde{F}'_{\text{diff}}\right)\right)\right). \tag{19}$$

E. Multitask Loss Function

In multitask learning, issues can arise such as significant differences in the gradient directions, convergence rates, and scales of loss across multiple tasks. Balancing these tasks to facilitate mutual improvement is a major challenge, and the loss function plays a critical role in this challenge.

The RSCD, in order to alleviate the problem of extremely imbalanced categories, uses a hybrid loss, i.e., combining the focal loss [57] and the dice loss [58], defined as follows:

$$L_{\text{cd}} = L_{\text{focal}} + L_{\text{dice}} \tag{20}$$

$$L_{\text{focal}} = -\alpha(1 - p_t)^{\gamma} \log(p_t) \tag{21}$$

$$L_{\text{dice}} = 1 - \frac{2 \cdot Y \cdot X}{Y + X} \tag{22}$$

where α represents a balance factor, which is used to balance the number of positive and negative samples. γ represents the focus parameter, which is used to adjust the model's focus on error-prone samples. In our experiments, α and γ are set to 1 and 0.5, respectively. p_t represents the probability of the model predicting correctly. When the prediction CMs is the same as the ground truth, $p_t = 1$, otherwise $p_t = 0$. Y represents the ground truth, and X represents the predicted CMs.

The RSCC uses cross-entropy loss function, which is defined as follows:

$$L_{\text{cc}} = -\sum_{i=1}^{n} \log\left(\sum_{j=1}^{m} Y_{i,j} X_{i,j}\right) \tag{23}$$

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where n is the sentence length and m is the vocabulary size. $Y_i = [Y_{i,1}, \ldots, Y_{i,j}, \ldots, Y_{i,m}]$ is the word vector representation at position i in the reference sentence. $X_i = [X_{i,1}, \ldots, X_{i,j}, \ldots, X_{i,m}]$ is a vector of word probability at position i in the predicted sentence, which is used to decide what word to put at this position.

To balance the learning speeds of the two tasks, we use a dynamic weighting method based on dynamic weight average (DWA) [37] combining L_{cd} and L_{cc}. DWA uses the change rate of loss at adjacent epochs to measure the learning rate of a task and to judge the degree of the task's convergence. And this is used to dynamically adjust the loss function weights of the two tasks, defining the weights of the tasks λ_k as

$$\lambda_k(t) := \frac{K \exp(w_k(t-1)/T)}{\sum_i \exp(w_i(t-1)/T)}$$
$$w_k(t-1) = \frac{L_k(t-1)}{L_k(t-2)} \tag{24}$$

where k, t are the task and iteration index, respectively. K denotes the number of tasks. $w_k(\cdot)$ denotes the relative rate of descent. T represents a temperature that controls the softness of task weighting. Finally, the softmax operator is multiplied by K to ensure that $\sum_i \lambda_i(t) = K$. $L_k(t)$ denotes the average loss over all iterations in an epoch.

However, the issue of inconsistent convergence speeds between tasks due to large differences in the magnitude of their losses still exists, leading to suboptimal performance of the model. To this end, we propose an enhanced DWA (EDWA) method. Based on the original DWA, we introduce a new mechanism to dynamically adjust the loss weights for smaller magnitude tasks, ensuring that their convergence rate matches that of larger magnitude tasks, thereby reducing model instability. Specifically, for the smaller magnitude task loss, we first normalize it and subsequently use the reciprocal of the larger magnitude task loss value as a moderating factor to scale this loss. In the way, when the losses of larger magnitude tasks are high, we can effectively suppress the contribution of the smaller magnitude task losses during backpropagation, thus enhancing the overall stability and performance of the model. The loss weight for smaller magnitude tasks is defined as

$$\alpha_s(t) = \frac{\lambda_k(t)}{L_l(t) L_s(t)} \tag{25}$$

where $L_l(t)$ and $L_s(t)$ denote the loss values of the larger and smaller magnitude tasks in the current iteration, respectively. The final overall loss function can be expressed as

$$L_{\text{all}} = \alpha_s L_{\text{cd}} + \lambda L_{\text{cc}}. \tag{26}$$

In this way, our method can dynamically balance the weights between different tasks during training, achieving better convergence and more stable model performance.

V. Experiments and Analysis

A. Implementation Details

The experiments are conducted under the Pytorch1.8 framework in Python3.6. Training and testing are completed on a workstation configured with an NVIDIA RTX 3050Ti GPU. The input image size is fixed to 256×256 and a pre-trained ResNet101 [49] is used as the backbone. The network is trained using the SGD optimizer (momentum $= 0.9$, weight decay $= 0.0005$) and initial learning rate is set to 0.01. To ensure fair comparison with the comparative methods, the rest of the hyperparameters are the same as those in the original paper.

B. Datasets and Evaluation Metrics

This section briefly describes the division of the datasets and the evaluation metrics for the RS-CDC task.

1) Dataset Setup: We verify the effectiveness of the CTMT-Net network on the two proposed datasets and one newly publicized LEVIR-MCI dataset, and the detailed information is described in Section III. Here, we only simply describe the division of the three datasets during the experiments. In the experiments, the LEVIR-CDC dataset is divided into training, validation, and testing sets, which include 6815, 1333, and 1929 pairs of images, respectively. The LEVIR-MCI dataset is also similarly divided. The WHU-CDC dataset is randomly divided into training, validation, and testing sets, which contain 5947, 743, and 744 pairs of images, respectively.

2) Evaluation Metrics for the CD Task: We use F1-score, intersection over union (IoU) of the change category, and overall accuracy (OA) as evaluation metrics for this task, with F1-score serving as the primary metric.

3) Evaluation Metrics for the CC Task: We use BLEU-N (N $= 1, 2, 3, 4$) [59], ROUGE-L [60], METEOR [61], and CIDEr-D [62] metrics to evaluate the accuracy of the captions. BLEU-N, ROUGE-L, METEOR, and CIDEr-D are all important evaluation metrics. BLEU-N is easily computable but ignores translation smoothness and semantic accuracy; ROUGE-L focuses more on translation smoothness and completeness but has higher computational complexity; METEOR can more accurately evaluate semantic accuracy; CIDEr-D metric is used to evaluate the IC tasks, considering the contextual information of words and the relevance of image content, which can more accurately measure semantic accuracy. The higher the score of these metrics, the more accurate the generated captions are.

C. Comparison to SOTA Methods

To evaluate the effectiveness of our method, a comprehensive performance evaluation was performed on the proposed LEVIR-CDC and WHU-CDC datasets. Currently, the methods that simultaneously handle CD and change captioning are lacking. Hence, we select some representative and open-source networks for comparison in RSCD and RSCC studies, respectively. For the task of RSCD, we choose SNUNet-CD [23], ChangeFormer [25], TFI-GR [63], A2Net [26], and HANet [64] as the comparison. Since these methods are all designed for the BCD tasks, we have adapted the labels of the LEVIR-MCI dataset in order to ensure fairness of comparison. Specifically, we did not use semantic labels with different colors to provide additional semantic information, but we uniformly use binary label of change/unchange to represent all building and road change categories. For the task

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TABLE IV

COMPARISON RESULTS ON THE LEVIR-CDC, WHU-CD, AND LEVIR-MCI DATASET. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%). THE BOLD REFERS TO THE OPTIMUM VALUES. THE UNDERLINE REFERS TO THE SECOND VALUES

		change detection			change captioning						
		F1	OA	IoU	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
LEVIR-CDC	**change detection**										
	SNUNet-CD(2021)	88.28	97.76	80.63	-	-	-	-	-	-	-
	ChangeFormer(2022)	92.02	98.46	86.04	-	-	-	-	-	-	-
	TFI-GR(2022)	88.49	98.87	79.36	-	-	-	-	-	-	-
	A2Net(2023)	90.51	**99.20**	82.66	-	-	-	-	-	-	-
	HANet(2023)	87.82	98.84	78.29	-	-	-	-	-	-	-
	change captioning										
	MCCFormers-S(2021)	-	-	-	79.90	70.26	62.68	56.68	36.17	69.46	120.39
	MCCFormers-D(2021)	-	-	-	80.42	70.87	62.86	56.38	37.29	70.32	124.44
	RSICCformer-C(2022)	-	-	-	83.09	74.32	66.66	60.44	38.76	72.63	130.00
	RSICCformer(2022)	-	-	-	84.72	76.27	68.87	62.77	**39.61**	74.12	134.12
	PSNet(2023)	-	-	-	83.86	75.13	67.89	62.11	38.80	73.60	132.62
	Prompt-CC(hard)(2023)	-	-	-	83.66	75.73	69.10	63.54	38.82	73.72	**136.44**
	Ours	**92.53**	98.54	**86.83**	**85.95**	**77.99**	**70.74**	**64.69**	39.49	**74.54**	134.94
WHU-CDC	**change detection**										
	SNUNet-CD(2021)	83.68	97.26	74.81	-	-	-	-	-	-	-
	ChangeFormer(2022)	86.11	98.01	77.88	-	-	-	-	-	-	-
	TFI-GR(2022)	87.39	98.94	77.60	-	-	-	-	-	-	-
	A2Net(2023)	**89.16**	98.52	80.43	-	-	-	-	-	-	-
	HANet(2023)	88.86	**99.02**	79.95	-	-	-	-	-	-	-
	change captioning										
	MCCFormers-S(2021)	-	-	-	82.14	76.29	71.08	66.51	43.50	**79.76**	148.88
	MCCFormers-D(2021)	-	-	-	73.29	67.88	64.03	60.96	39.69	73.67	134.92
	RSICCformer-C(2022)	-	-	-	78.25	72.82	68.57	65.14	44.35	76.50	143.44
	RSICCformer(2022)	-	-	-	80.05	74.24	69.61	66.54	42.65	73.91	133.44
	Ours	88.93	98.01	**81.54**	**83.56**	**77.66**	**72.76**	**69.00**	**45.39**	79.23	**149.40**
LEVIR-MCI	**change detection**										
	SNUNet-CD(2021)	89.79	96.70	82.53	-	-	-	-	-	-	-
	ChangeFormer(2022)	90.33	96.96	83.35	-	-	-	-	-	-	-
	TFI-GR(2022)	88.48	98.02	79.34	-	-	-	-	-	-	-
	A2Net(2023)	89.39	**98.04**	80.82	-	-	-	-	-	-	-
	HANet(2023)	83.20	97.13	71.23	-	-	-	-	-	-	-
	change captioning										
	MCCFormers-S(2021)	-	-	-	79.90	70.26	62.68	56.68	36.17	69.46	120.39
	MCCFormers-D(2021)	-	-	-	80.42	70.87	62.86	56.38	37.29	70.32	124.44
	RSICCformer-C(2022)	-	-	-	83.09	74.32	66.66	60.44	38.76	72.63	130.00
	RSICCformer(2022)	-	-	-	84.72	76.27	68.87	62.77	**39.61**	**74.12**	134.12
	PSNet(2023)	-	-	-	83.86	75.13	67.89	62.11	38.80	73.60	132.62
	Prompt-CC(hard)(2023)	-	-	-	83.66	75.73	69.10	63.54	38.82	73.72	**136.44**
	Ours	**91.67**	97.38	**85.37**	**85.16**	**77.24**	**70.13**	**64.16**	39.34	74.03	134.52

of RSCC, we choose MCCFormers-S [54], MCCFormers-D [54], RSICCformer-C [33], RSICCformer [33], PSNet [34], and Prompt-CC(hard) [36] as the comparison, in which the last two works are not open source. As a result, we are unable to reproduce the results of the work on the WHU-CDC dataset.

1) Quantitative Comparison: The results of the different methods on the two proposed datasets are presented in Table IV. The table provides a complete comparison of all the metrics on which we can analyze the different models. Overall, the CTMTNet achieves two tasks while having better performance than other methods. For example, on LEVIR-CDC dataset, our method achieves the highest in F1, IoU, BLEU-1, BLEU-2, BLEU-3, BLEU-4, and ROUGE-L metrics, with values that are 0.51%, 0.79%, 1.23%, 1.72%, 1.64%, 1.15%, and 0.42% higher than the second place, respectively. On the WHU-CDC dataset, our method ranks first in IoU, BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, and CIDEr-D metrics, with values that are 1.11%, 1.42%, 1.37%, 1.68%, 2.46% 1.04%, and 0.52% higher than the second place, respectively. This show that our proposed method can simultaneously fulfill the RSCD and RSCC task requirements.

To further demonstrate the effectiveness of our model, we also performed a comprehensive performance evaluation on a newly publicized LEVIR-MCI dataset, as shown in Table IV. In particular, for the single RSCC task, the image pairs and caption labels of the LEVIR-MCI and LEVIR-CDC datasets are identical, equivalent to the LEVIR-CC dataset. Therefore, the metric values of the comparison methods on LEVIR-MCI are the same as LEVIR-CDC. For the RSCD task, compared to the LEVIR-CDC dataset (labeled with single-category changes of buildings), the performance of some comparative methods is significantly decreased on the LEVIR-MCI dataset (labeled with multicategory changes in buildings and roads), such as ChangeFormer, A2Net, and HANet. Our method exhibits better robustness, thanks to our model's ability to handle both the RSCD and RSCC tasks simultaneously. The complementary use of positional and semantic information makes the model perform more stably across different datasets.

2) Qualitative Comparison: To further illustrated the superiority of CTMTNet, we perform visualization and analysis on the two proposed datasets. Figs. 8 and 9 show the a visual comparison of the binary CMs generated by the CD task on LEVIR-CDC and WHU-CDC datasets. Macroscopically, on the LEVIR-CDC dataset, the TFI-GR, ANet, HANet, and the proposed CTMTNet methods are generally effective. But on the WHU-CDC dataset, some methods are poorly effective,

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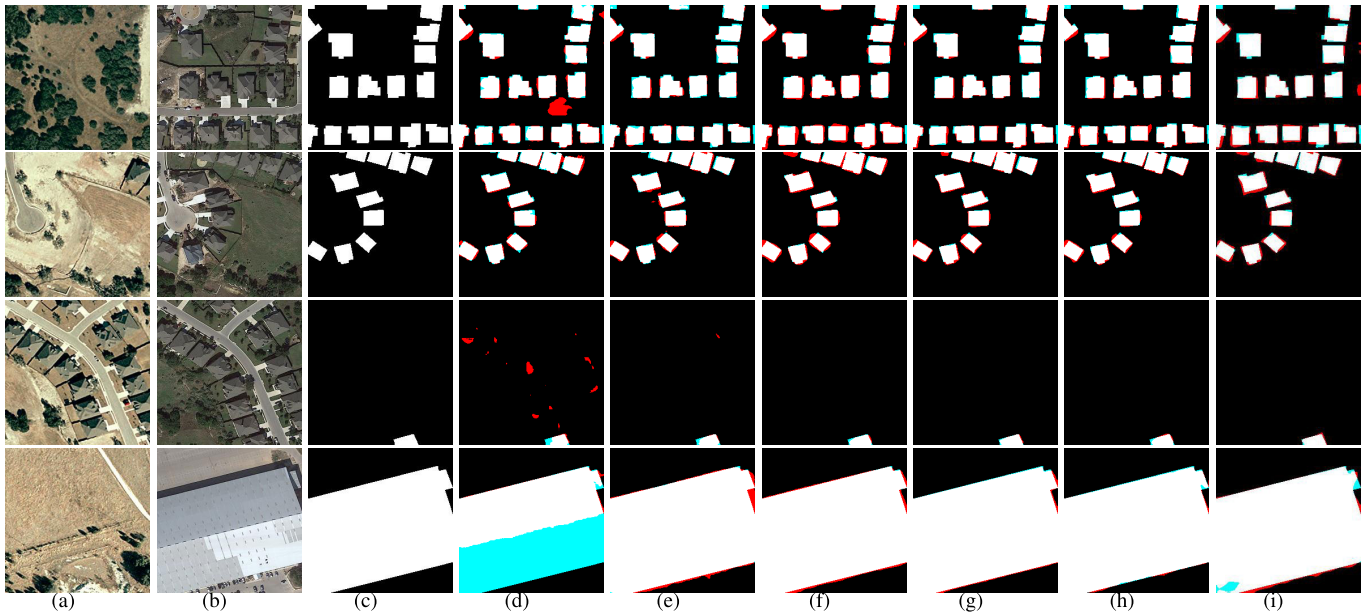


Fig. 8. CD results of the LEVIR-CDC dataset. (a) Pre-image. (b) Post-image. (c) Ground truth. (d) SNUNet-CD. (e) ChangeFormer. (f) TFI-GR. (g) A2Net. (h) HANet. (i) CTMTNet(ours). True positives (TP) are represented by white pixels, while true negatives (TN) are black. False positives (FP) are red and false negatives (FN) are blue.

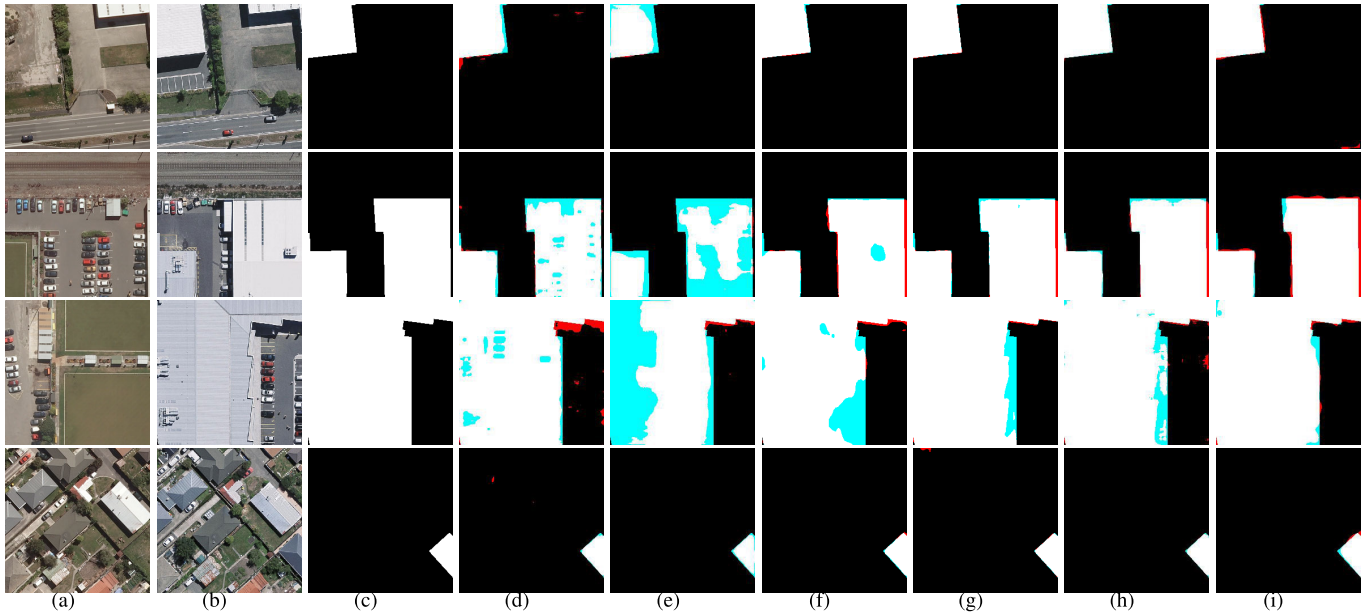


Fig. 9. CD results of the WHU-CDC dataset. (a) Pre-image. (b) Post-image. (c) Ground truth. (d) SNUNet-CD. (e) ChangeFormer. (f) TFI-GR. (g) A2Net. (h) HANet. (i) CTMTNet (ours). True positives (TP) are represented by white pixels, while true negatives (TN) are black. False positives (FP) are red and false negatives (FN) are blue.

with more false negatives and higher error rates. Especially concerning the change of large buildings, as in rows 2 and 3 of Fig. 9, the scene changes from parking lots, grassy areas and roads to large buildings. From the visualization results, most methods exhibit incomplete detection and miss edges. Our proposed method can effectively detect large building changes, maintain edge integrity, and reduce error rates.

We also visualize the change captions generated by the CC task on the LEVIR-CDC and WHU-CDC datasets, respectively, as shown in Figs. 10 and 11. We provide one of the five annotated sentences for each image pair and show the captions generated by the comparison method and our proposed method. We highlight inaccurate descriptions and misdescriptions in red and more accurate descriptions in blue. It is observed that our model is able to generate more accurate captions than the comparison methods in complex scenes. On the LEVIR-CDC dataset, our model do not generate redundant words, e.g., the first, second, and fourth picture pairs, and the sentences generated by the MCCFormers-S and RSICCformer-C methods would have redundant words that resulted in erroneous predictions. In the WHU-CDC dataset, the first, second, and third image pairs are all recognized as

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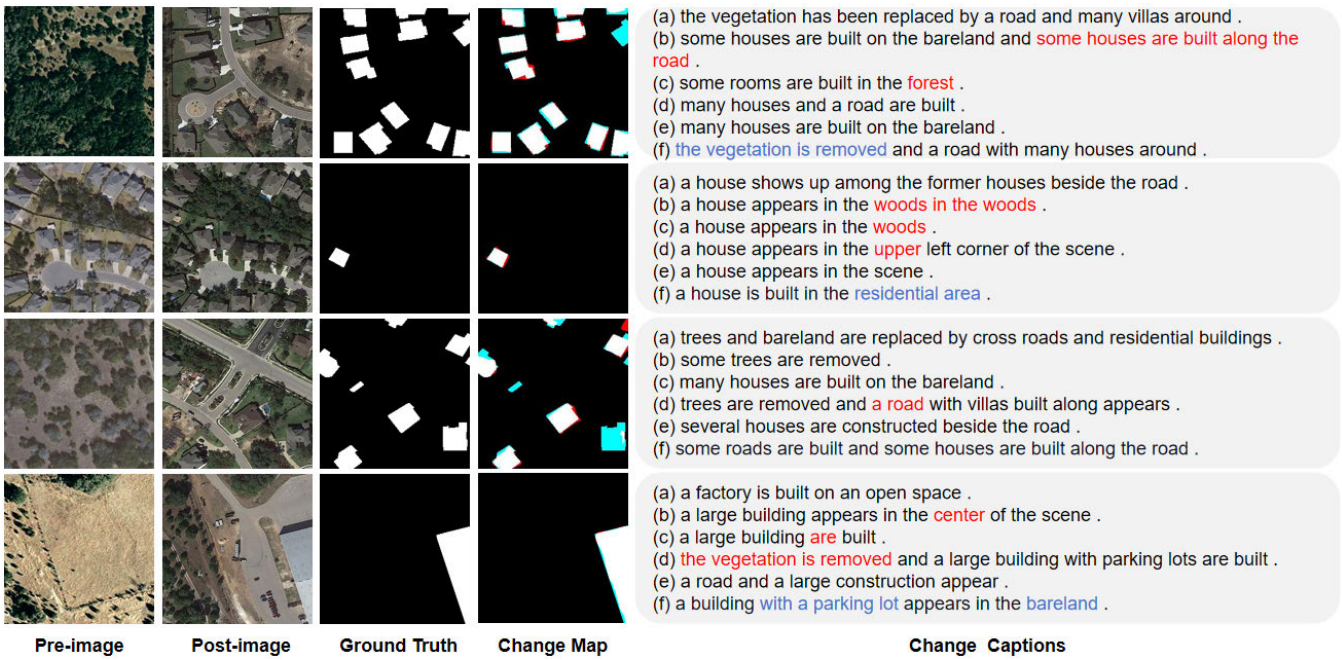


Fig. 10. Captioning results on the LEVIR-CDC dataset. (a) Reference sentence. (b) MCCFormers-S. (c) MCCFormers-D. (d) RSICCformer-C. (e) RSICCformer. (f) CTMTNet (Ours).



Fig. 11. Captioning results on the WHU-CDC dataset. (a) Reference sentence. (b) MCCFormers-S. (c) MCCFormers-D. (d) RSICCformer-C. (e) RSICCformer. (f) CTMTNet (Ours).

identical by the MCCFormers-D method. In the second image pair, the RSICCformer-C method predictions only contain the objects in the image at the latter moment, without describing what has changed. In the third image pair, MCCFormers-S and RSICCformer-C cannot accurately predict the location and number information of the changing objects, respectively. And our method shows more accurate prediction results, such as "top."

The above experimental results show that our CTMTNet can improve the performance of RSCD and RSCC tasks effectively. The CTMTNet could accurately locate the position of the change object, identify the relevant changes, and generate accurate sentences describing the changes.

D. Parameter Analysis

The depth of the MAFEM module has a great impact on the model's performance. In this section, we evaluate the impact of MAFEM performance on experimental metrics for different depths on the LEVIR-CDC dataset. Table V

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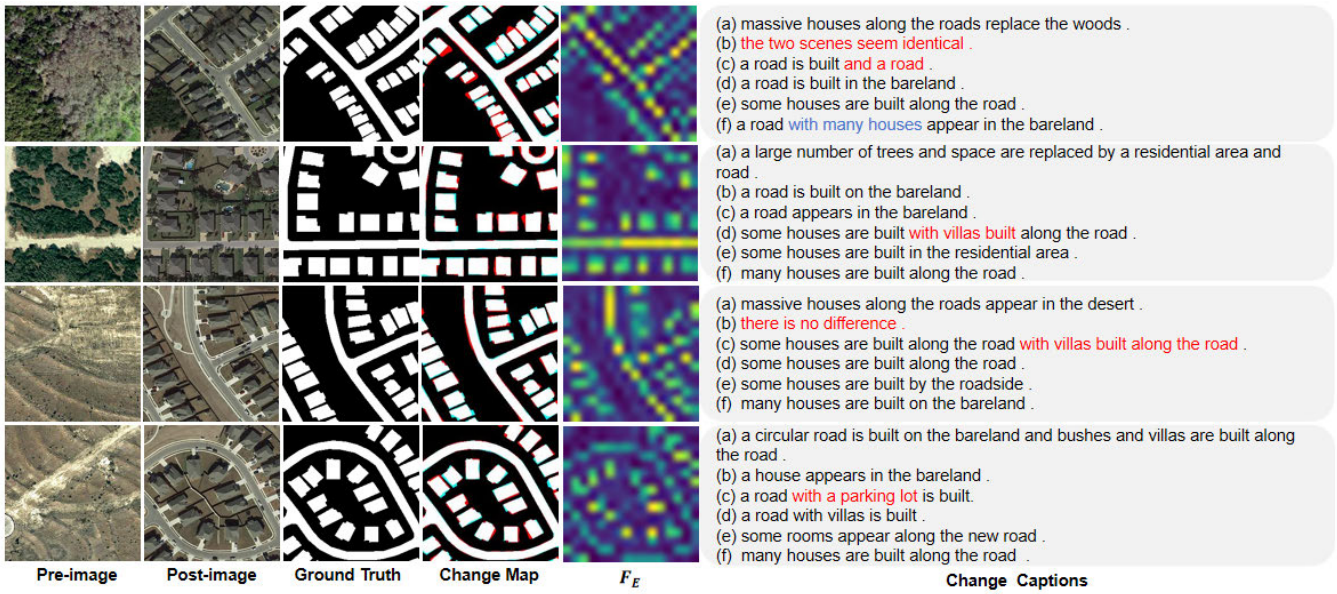


Fig. 12. Visualization of image embedding and change captioning results on LEVIR-MCI dataset. F_E denotes image embedding. (a) Reference sentence. (b) MCCFormers-S. (c) MCCFormers-D. (d) RSICCformer-C. (e) RSICCformer. (f) Predicted sentence.

TABLE V
ANALYSIS OF THE NUMBER OF LAYERS IN MAFEM MODULE. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%). THE BOLD REFERS TO THE OPTIMUM VALUES

MAFEM layer	F1	OA	IoU	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
1	**93.02**	**98.61**	**87.59**	84.86	76.45	68.95	62.92	38.90	73.53	131.91
2	92.53	98.54	86.83	**85.95**	**77.99**	**70.74**	64.69	39.49	74.54	**134.94**
3	92.68	98.56	87.06	85.64	77.56	70.22	64.21	**39.65**	**74.68**	134.50
4	92.83	98.57	87.29	85.26	77.62	70.65	**64.94**	39.36	74.39	134.71
5	92.68	98.54	87.05	85.25	76.70	69.03	62.72	38.99	73.92	133.02

shows the quantitative results, where MAFEM layer represents the depth of MAFEM. We can observe that the proposed CTMTNet model achieves the best performance when the depth of MAFEM is 2.

E. Ablation Study

On the LEVIR-CDC dataset, we perform ablation experiments for the MAFEM and FFB modules to verify the effectiveness of each part of the CTMTNet. These include four experiments as follows.

1) *Baseline:* This experiment only uses the ResNet101 backbone and multitask decoder.
2) *FFB:* This experiment adds FFB to the baseline.
3) *MAFEM:* This experiment adds MAFEM to the baseline.
4) *CTMTNet:* This is the proposed CTMTNet.

Table VI provides the results of the ablation experiments. From the results on the LEVIR-CDC dataset, it can be seen that when adding the FFB module to fuse bitemporal image deep semantic features based on the baseline, all metrics were improved except for the IoU metric which decreased. When adding the MAFEM module to enhance image features based on the baseline, it can improve all metrics, among which the BLEU-2, BLEU-3, BLEU-4, and CIDEr-D metrics are significantly improved by 2.47%, 2.6%, 2.73%, and 6.73%, respectively. When both MAFEM and FFB modules are introduced, it can be observed that the performance is significantly improved compared to the baseline, only adding the MAFEM or FFB modules. From the results of ablation experiments, the proposed MAFEM and FFB modules are effective and can significantly improve the performance of our dual-task networks. In summary, these results confirm that our method is effective and has the potential to advance the RS-CDC task field.

F. Qualitative Visualization

To evaluate the ability of our proposed CTMTNet method for accurately identify relevant changes. We use the LEVIR-MCI dataset as an example selecting scenes that contain both road and building change categories for evaluation. We visualize the image embeddings F_E extracted from the FEFM module and the CMs and captions generated by the two decoders, respectively, as shown in Fig. 12. In particular, for a clear representation of the accuracy of the predicted CMs, we have labeled the correctly predicted road and building categories in white, false positives in red, and false negatives in blue. We can see that the visualized image embedding demonstrates that the proposed FEFM module could accurately identify the relevant changes in the bitemporal images. On the basis of this, the two decoding branches are able to accurately generate CMs and change captions.

G. Analysis of Loss Function

This section analyzes the impact of the loss function and loss weighting method for each task (RSCD and RSCC) in

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TABLE VI

ANALYSIS OF ABLATION EXPERIMENTS ON MAFEM AND FFB MODULES. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%). THE BOLD REFERS TO THE OPTIMUM VALUES

MAFEM	FFB	F1	OA	IoU	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
✘	✘	89.77	98.09	83.87	82.93	74.53	67.45	61.52	37.46	71.76	124.38
✘	✔	90.58	98.12	82.72	83.14	75.61	69.21	64.04	37.64	72.36	126.78
✔	✘	91.46	98.33	85.19	84.76	77.00	70.05	64.25	38.89	73.71	131.11
✔	✔	**92.53**	**98.54**	**86.83**	**85.95**	**77.99**	**70.74**	**64.69**	**39.49**	**74.54**	**134.94**

TABLE VII

ANALYSIS OF LOSS FUNCTION. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%). THE BOLD REFERS TO THE OPTIMUM VALUES

method	L_{cd}	L_{cc}	F1	OA	IoU	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D
MANUAL	CE	CE	91.89	98.36	85.83	82.75	73.75	66.25	60.21	37.89	72.16	127.89
MANUAL	focal+dice	CE	90.83	98.15	84.24	78.96	70.42	62.48	55.09	35.26	72.76	126.94
UNCERT	CE	CE	**92.90**	**98.58**	**87.40**	81.67	72.66	65.05	59.04	36.22	70.51	122.64
UNCERT	focal+dice	CE	89.99	98.02	83.02	79.87	71.93	65.37	60.94	35.86	70.21	120.67
DWA	CE	CE	91.65	98.33	85.47	85.09	77.16	69.40	62.97	39.13	74.27	131.92
DWA	focal+dice	CE	92.37	98.49	86.58	85.25	77.58	70.14	63.93	**39.56**	**74.69**	133.06
EDWA	CE	CE	92.76	98.57	87.19	85.70	77.00	69.22	62.88	39.47	74.34	133.58
EDWA	focal+dice	CE	92.53	98.54	86.83	**85.95**	**77.99**	**70.74**	**64.69**	39.49	74.54	**134.94**

TABLE VIII

COMPARATIVE ANALYSIS OF SINGLE-TASK LEARNING AND MULTITASK LEARNING. PARAMS AND TIME METRICS INDICATE THE NUMBER OF PARAMETERS IN THE MODEL AND THE INFERENCE TIME ON THE TEST SET, RESPECTIVELY. EXCEPT FOR THE PARAMS AND TIME METRICS, THE OTHER SCORES ARE DESCRIBED IN PERCENTAGE (%). THE BOLD REFERS TO THE OPTIMUM VALUES

method	F1	OA	IoU	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	param(M)	time(s)
CD task	90.93	98.24	84.40	-	-	-	-	-	-	-	36.83	93.29
CC task	-	-	-	82.53	74.00	66.55	60.36	37.48	71.22	125.71	73.37	134.15
Multitask	**92.53**	**98.54**	**86.83**	**85.95**	**77.99**	**70.74**	**64.69**	**39.49**	**74.54**	**134.94**	73.53	133.72

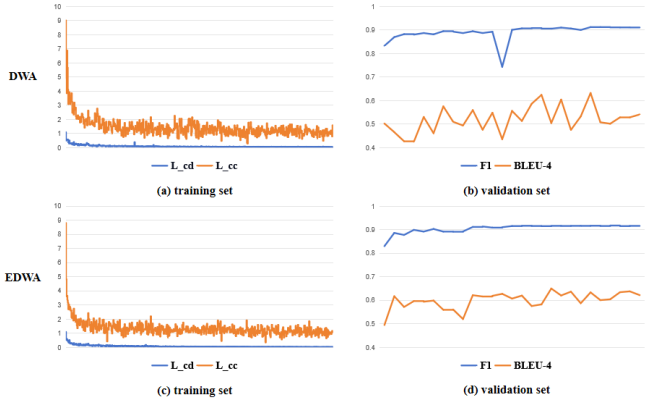


Fig. 13. Loss changes during training and metric changes on the validation set for the DWA and EDWA methods on the LEVIR-CDC dataset. (a) and (c) Training set. (b) and (d) Validation set.

the CTMTNet model. Uncert [56] already demonstrated that model performance is sensitive to the weighting of losses, and correctly weighting the loss functions of different tasks is crucial for multitask learning.

In the application of DWA [37], we noticed that due to the magnitude difference between the two tasks, there is inconsistent convergence speed, leading to noticeable instability in model performance, particularly for the RSCC task. As shown in Fig. 13, we visualized the loss changes during the training process and the performance metrics changes on the validation set (F1 and BLEU-4 are used as representative metrics for the RSCD and RSCC tasks, respectively). As can be seen from (a) and (b), the RSCC task's performance metrics fluctuate significantly, leading to a negative impact on the performance of the RSCD task as well. In particular, in the middle epoch, the performance metrics of both tasks show a dramatic decrease. (c) and (d) are the results using our EDWA weighting method, and it can be observed that the convergence rate of the two tasks is more consistent. On the validation set, the RSCC task's performance metrics are relatively stable, while the RSCD task's metrics show a steady increase.

We also compare the experimental results for different combinations of weighting strategies and loss functions, as shown in Table VII. When using manual weighting (MANUAL), both tasks are set to 0.5. When using the Uncert method [41] and CE loss, the RSCD task achieves peak performance, but the RSCC task does not perform well and fails to achieve a balance between the two tasks. When using the DWA method, it can be observed that no matter what loss function is used, the performance of both tasks could been well balanced. And when we use the EDWA method, the performance on both tasks is better than the case of using the DWA method. These results further demonstrate that our method can better balance the two tasks during training, leading to more stable model performance.

In addition, we also conducted comparative experiments between single-task and multitask, and the single-task experiments include CD (using a mixed focal and dice loss) and CC (using CE loss) tasks, respectively. As shown in Table VIII, compared to single-task, the metrics generated using multitask learning are significantly improved, except for the OA metrics of the CD task. We also counted the number of parameters and the inference time on the test set for single-task and multitask networks. It can be seen that the parameter count of the

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multitask network is much smaller than the sum of parameters for the two independent tasks, primarily due to our adoption of a shared lower-level network structure. In addition, the lightweight CD decoder brings only a small number of parameters, so the multitask parameter count is only 0.16 M more than the CC task. On the other hand, the real inference time of the model depends not only on the number of parameters but is also affected by multiple factors such as memory access patterns and network architecture. For the complex CC task, the simple CD task is introduced as an implicit regularization means to enable the model to learn more common feature representations and reduce the risk of overfitting. Thus, compared to the CC task, the dual-task model shows higher efficiency in inference, despite the increase in the number of parameters. In summary, our method is able to obtain richer and more detailed interpretation results in a shorter inference time. It is further demonstrated that multitask learning is beneficial to improve the performance of single task. Under the one framework, two tasks can be realized concisely and efficiently at the same time to obtain richer interpretation results and satisfy more application scenarios. This significantly saves time and reduces the workload compared to handling the two tasks independently.

VI. Discussion and Conclusion

In this article, we simply and efficiently combine the RSCD task and the CC task (RS-CDC). And the RS-CDC task is efficiently solved in a multitasking framework [i.e., the new multitasking network (CTMTNet)], which effectively utilizes the complementary information of the two tasks to further enhance the stability of the model. To verify our method, we built two large-scale RS-CDC datasets, named LEVIR-CDC dataset and WHU-CDC dataset, respectively. In CTMTNet, we extract visual features of bitemporal images through a Siamese CNN backbone network and design MAFEM and FFB modules to enhance and fuse the bitemporal features, respectively. Finally, two decoding branches are used to output binary CMs that locate the changed objects and sentences that describe the changes, respectively. During the training phase, we adopted an improved dynamic weighting method to weight the losses of the two tasks, deriving the total loss function to guide model training. This method makes the overall performance of the model more stable. Experimental results on the two proposed datasets and a newly publicized LEVIR-MCI dataset demonstrate that our model is effective and can achieve SOTA performance. Our proposed method is able to achieve a good balance between RSCD and RSCC tasks.

Nevertheless, it should be acknowledged that there is still space for further enhancement of our work, such as reducing the number of parameters and model training time. In addition, the current training requires labels for both RSCD and RSCC tasks, which are not easy to obtain and require a lot of labor and time. Finally, our future work aims to research automated annotation tools for RSCC task labels to reduce labor and time consumption and provide more data support for advancing the RS-CDC task. We will also improve our approach to address some existing problems with the current model.

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Jingye Shi received the bachelor's and master's degrees from the School of Computer and Information Technology, Beijing Jiaotong University, Beijing, China, in 2005 and 2008, respectively, where he is currently pursuing the Ph.D. degree with Beijing Key Laboratory of Security and Privacy in Intelligent Transportation.

His research interests include signal processing and intelligent technologies in remote sensing image processing.

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Mengge Zhang received the bachelor's degree in computer and information technology from Nanyang Normal University, Nanyang, Henan, China, in 2020. She is currently pursuing the master's degree with the National Supercomputing Center and the School of Computer and Artificial Intelligence, University of Zhengzhou, Zhengzhou, China.

Her research interests include computer vision, change detection, and change captioning in remote-sensing images.



Yuewu Hou received the bachelor's degree in software engineering from Changchun Institute of Technology, Changchun, China, in 2019, and the master's degree in computer science and engineering from Beijing University of Technology, Beijing, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Computer and Communication Engineering, University of Science and Technology Beijing, Beijing.

His research interests include computer vision and multimodal knowledge graphs.



Ruicong Zhi received the Ph.D. degree in signal and information processing from Beijing Jiaotong University, Beijing, China, in 2010.

She visited the KTH Royal Institute of Technology, Stockholm, Sweden, in 2008, as a joint Ph.D. Student. From 2016 to 2017, she visited the University of South Florida, Tampa, FL, USA, as a Visiting Scholar. She is currently a Professor with the School of Computer and Communication Engineering, University of Science and Technology Beijing, Beijing. She has published more than 80 articles and more than 20 patents. Her research interests include artificial intelligence and signal processing.



Jiqiang Liu received the B.S. and Ph.D. degrees from Beijing Normal University, Beijing, China, in 1994 and 1999, respectively.

He is currently a Professor at the School of Computer and Information Technology, Beijing Jiaotong University, Beijing. He has published over 200 scientific papers in various journals and international conferences. His main research interests are privacy-preserving, Internet of Things (IoT) security, and cryptographic protocols.

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